

Innovative modelling tools for strengthening global health resilience

Martín Lotto Batista, PhD
Barcelona Supercomputing Center (BSC)



**Climate-Sensitive Vector Dynamics
Modelling Workshop**
18 September 2025 - Nicosia, Cyprus

COMPUTER SCIENCES



Prof Rachel Lowe
Leading Researcher

Martín Lotto Batista, MSc
Research Engineer

Global Health Resilience (GHR)

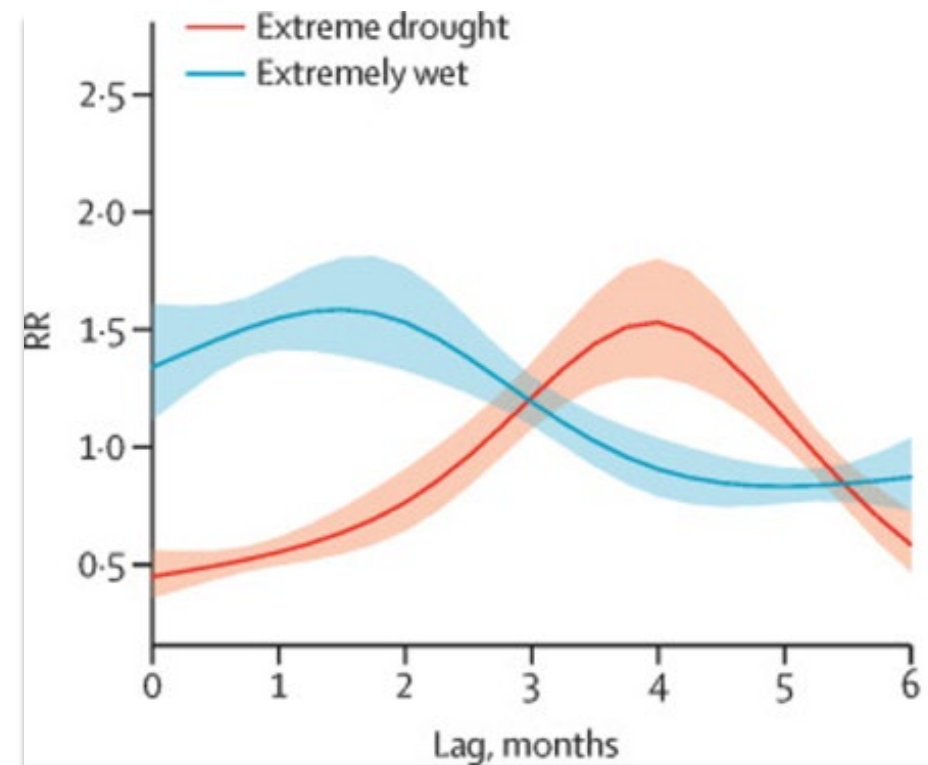


Climate-based CSID models

Mathematical formulations of the relationship between climate and disease

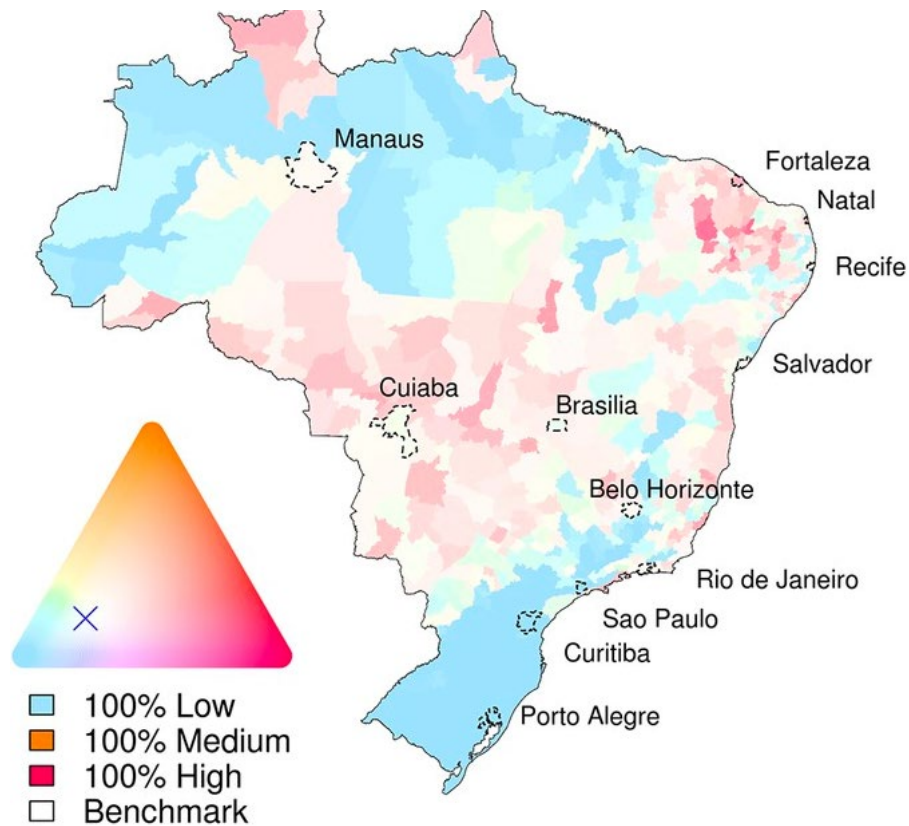
Goals

- Identify risk factors
- Develop decision-support tools
- Detect spatial or temporal trends
- Identify historical and future trends

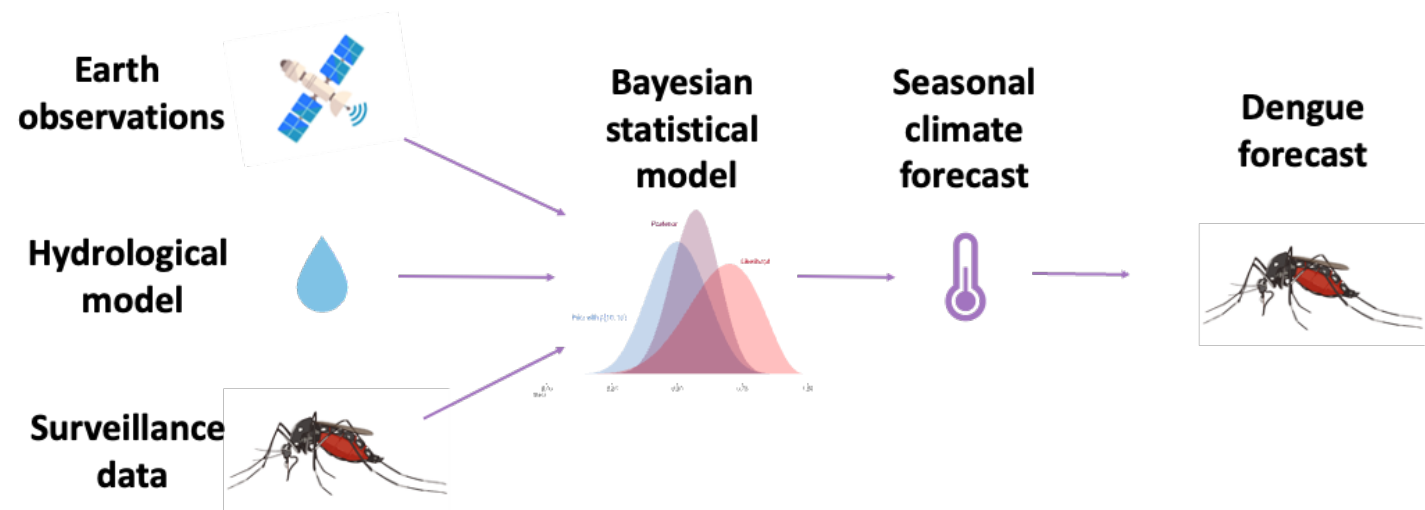


Lowe et al., 2021

Climate-based CSID models



In several of our projects (e.g. [E4Warning](#), [IDAlert](#), [IDExtremes](#)) we develop infectious disease **forecast system models** in South America, Europe and Asia.



We are exploring how new data streams and finer resolution data can improve forecasting models

Climate-based CSID models

THE LANCET
Planetary Health

ARTICLES | VOLUME 1, ISSUE 4, E142-E151, JULY 2017

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Climate services for health: predicting the evolution of the 2016 dengue season in Machala, Ecuador

Dr Rachel Lowe, PhD  • [Anna M Stewart-Ibarra, PhD](#) • [Desislava Petrova, MSc](#) • [Markel García-Díez, PhD](#) • [Mercy J Borbor-Cordova, PhD](#) • [Raúl Mejía, MSc](#) • et al. [Show all authors](#)

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Research articles

Towards a leptospirosis early warning system in northeastern Argentina

Martín Lotto Batista[†], Eleanor M. Rees[†], Andrea Gómez, Soledad López, Stefanie Castell, Adam J. Kucharski, Stéphane Ghazzi, Gabriela V. Müller and Rachel Lowe 

Published: 17 May 2023 | <https://doi.org/10.1098/rsif.2023.0069>

PLOS MEDICINE

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RESEARCH ARTICLE

Nonlinear and delayed impacts of climate on dengue risk in Barbados: A modelling study

Rachel Lowe , Antonio Gasparini, Cédric J. Van Meerbeeck, Catherine A. Lippi, Roché Mahon, Adrian R. Trotman, Leslie Rollock, Avert O. J. Hinds, Sadie J. Rvan, Anna M. Stewart-Ibarra

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Combined effects of hydrometeorological hazards and urbanisation on dengue risk in Brazil: a spatiotemporal modelling study

Rachel Lowe, PhD  • [Sophie A Lee, MSc](#) • [Kathleen M O'Reilly, PhD](#) • [Oliver J Brady, DPhil](#) • [Leonardo Bastos, PhD](#) • [Gabriel Carrasco-Escobar, MSc](#) • et al. [Show all authors](#)

Open Access • Published: April, 2021 • DOI: [https://doi.org/10.1016/S2542-5196\(20\)30292-8](https://doi.org/10.1016/S2542-5196(20)30292-8) •  Check for updates

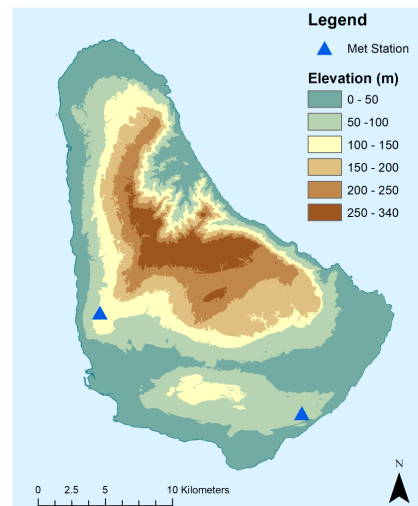
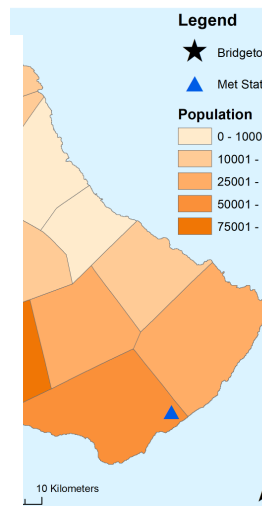
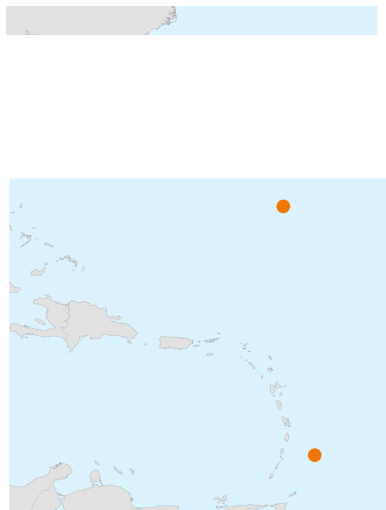
Interactions between climate change, urban infrastructure and mobility are driving dengue emergence in Vietnam

[Rory Gibb](#) , [Felipe J. Colón-González](#), [Phan Trong Lan](#), [Phan Thi Huong](#), [Vu Sinh Nam](#), [Vu Trong Duoc](#), [Hung Do Thai](#), [Nguyễn Thanh Dong](#), [Vien Chinh Chien](#), [Ly Thi Thuy Trang](#), [Do Kien Quoc](#), [Tran Minh Hoa](#), [Nguyen Hữu Tai](#), [Tran Thi Hang](#), [Gina Tsarouchi](#), [Eleanor Ainscoe](#), [Quillon Harpham](#), [Barbara Hofmann](#), [Darren Lumbroso](#), [Oliver J. Brady](#) & [Rachel Lowe](#)

[Nature Communications](#) **14**, Article number: 8179 (2023) | [Cite this article](#)

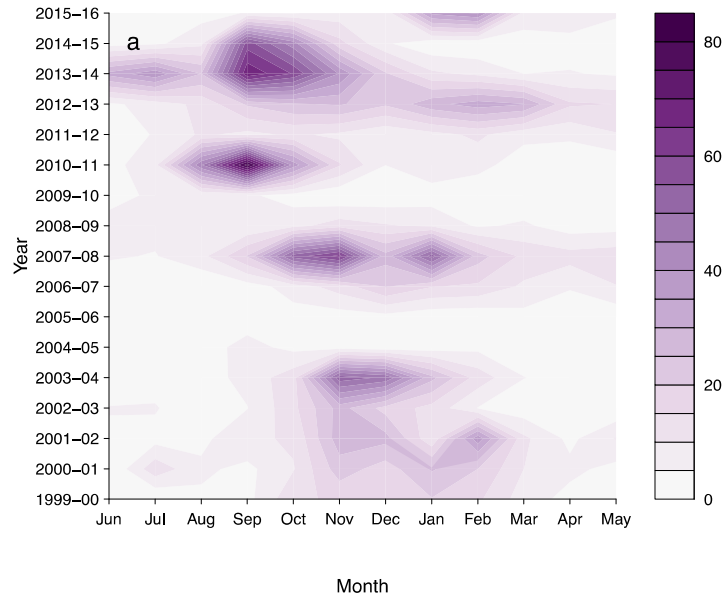
Arboviruses in the Caribbean

- The Caribbean region is facing a major crisis of co-occurring epidemics of **dengue**, chikungunya and Zika viruses.
- An increase incidence of dengue suspected in **Barbados** following **drought** events.

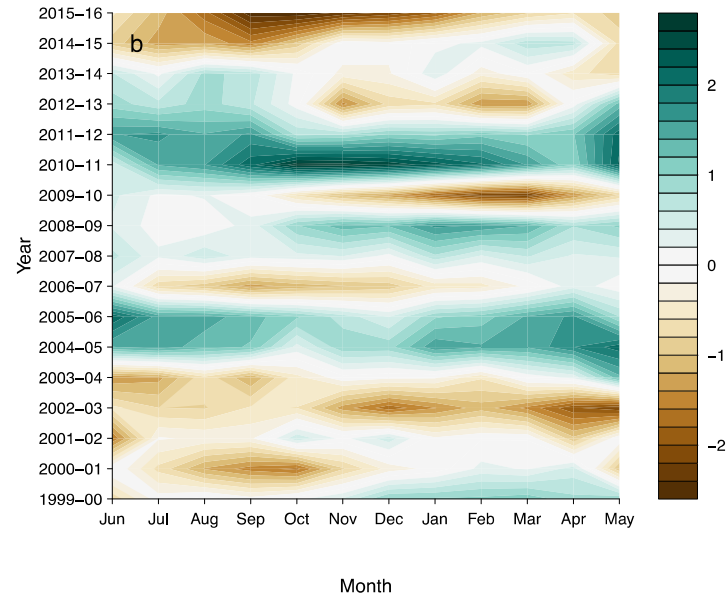


Dengue and hydrometeorological indicators

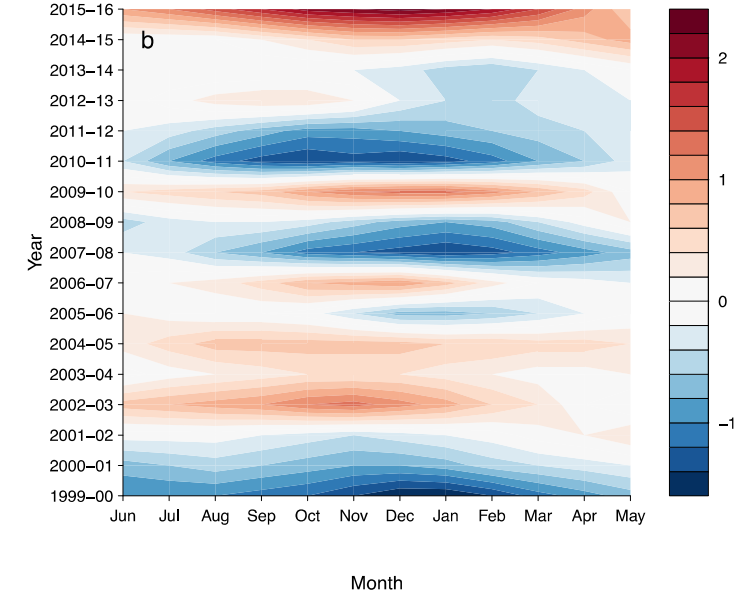
Dengue incidence rate



Standardised precipitation index

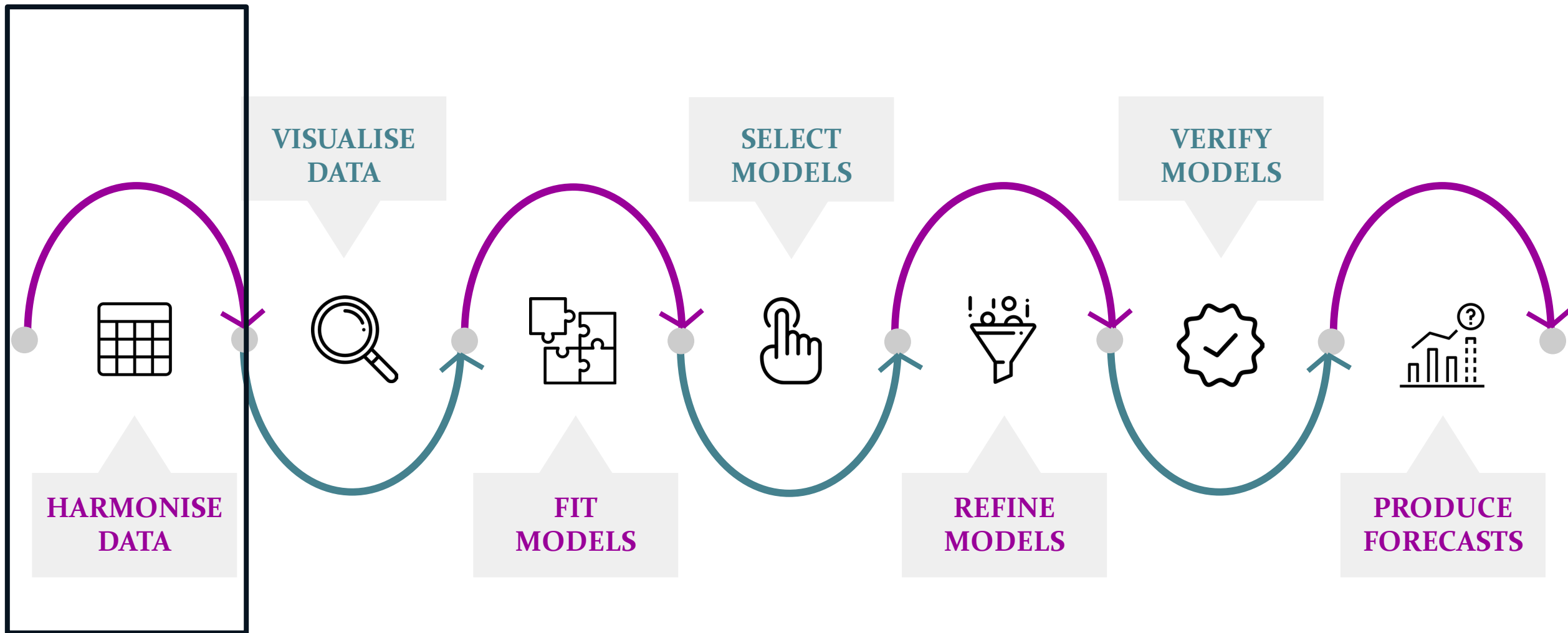


Oceanic Niño index



We used **distributed lag non-linear models** (Gasparrini, 2013) coupled with a **Bayesian hierarchal model** (Lowe et al, 2017) to understand exposure-lag response associations between dengue, temperature, drought and rainfall.

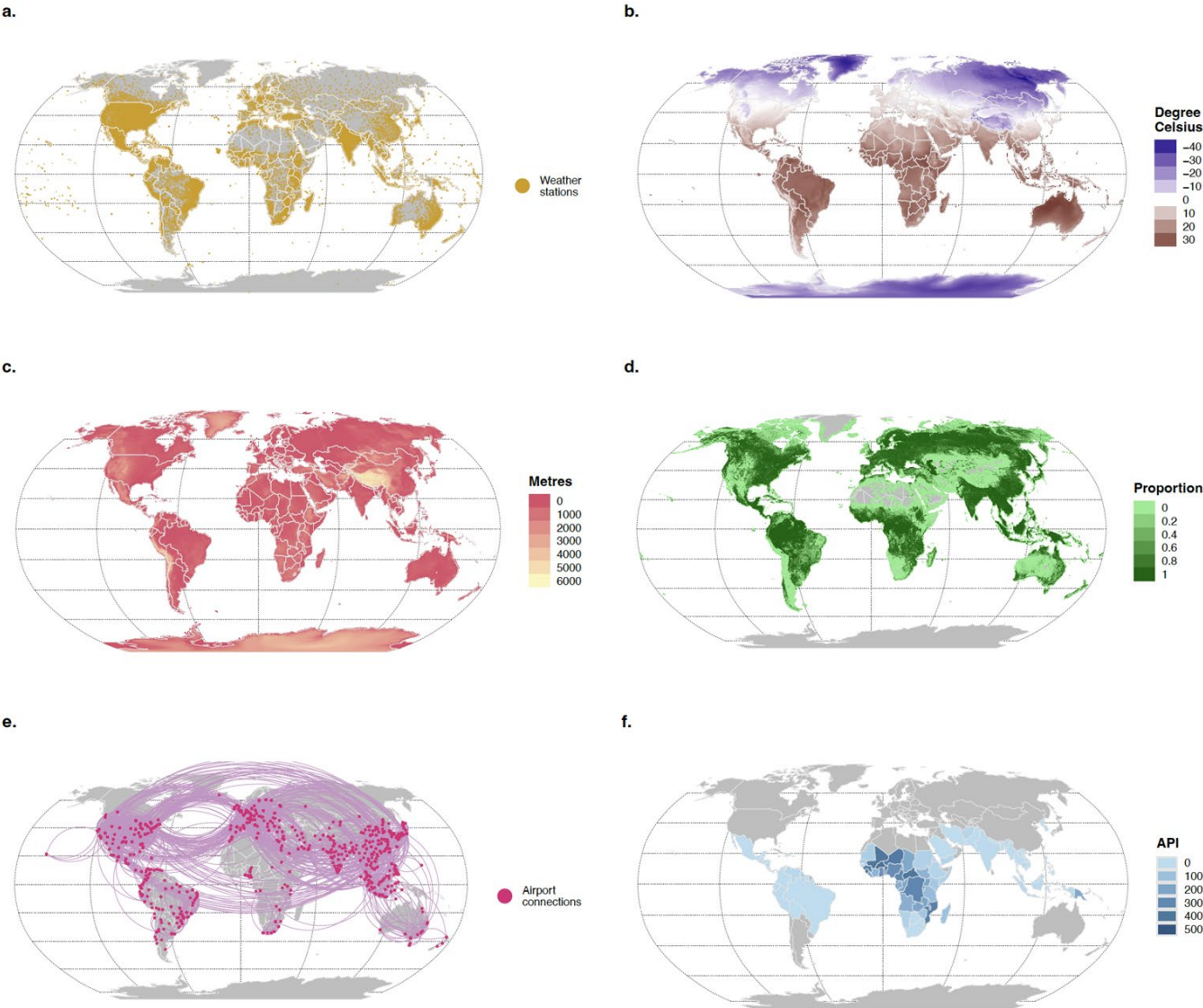
Risk modelling



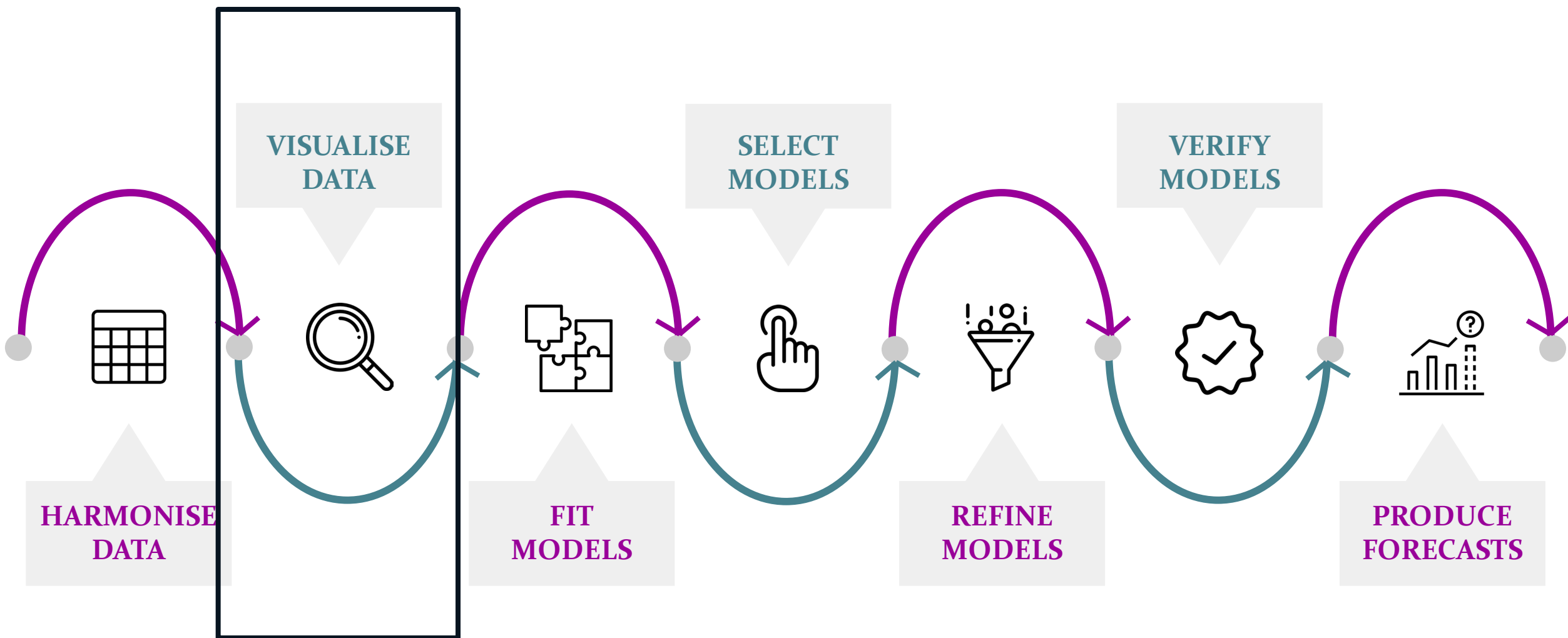
Data harmonisation



HARMONIZE

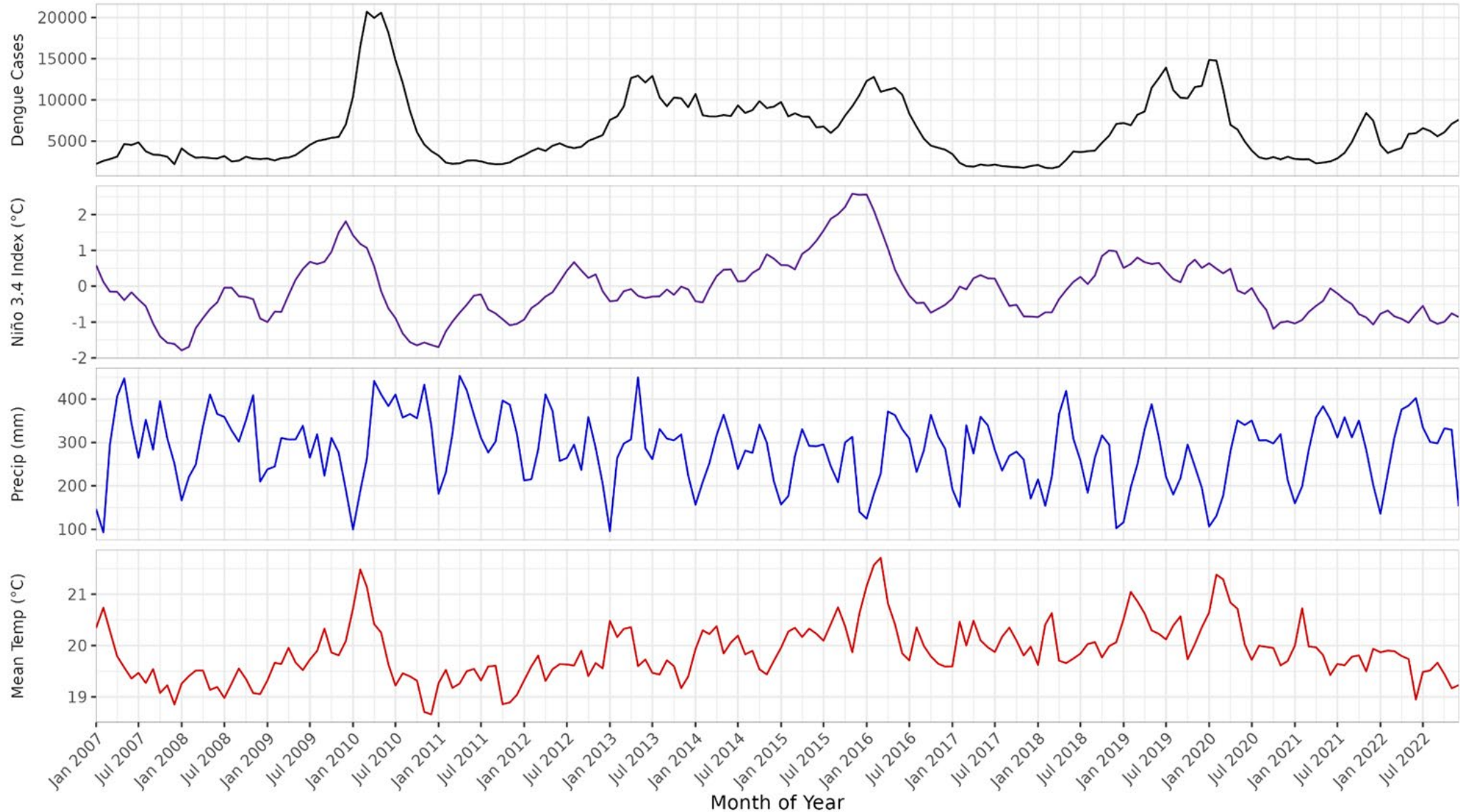


Risk modelling



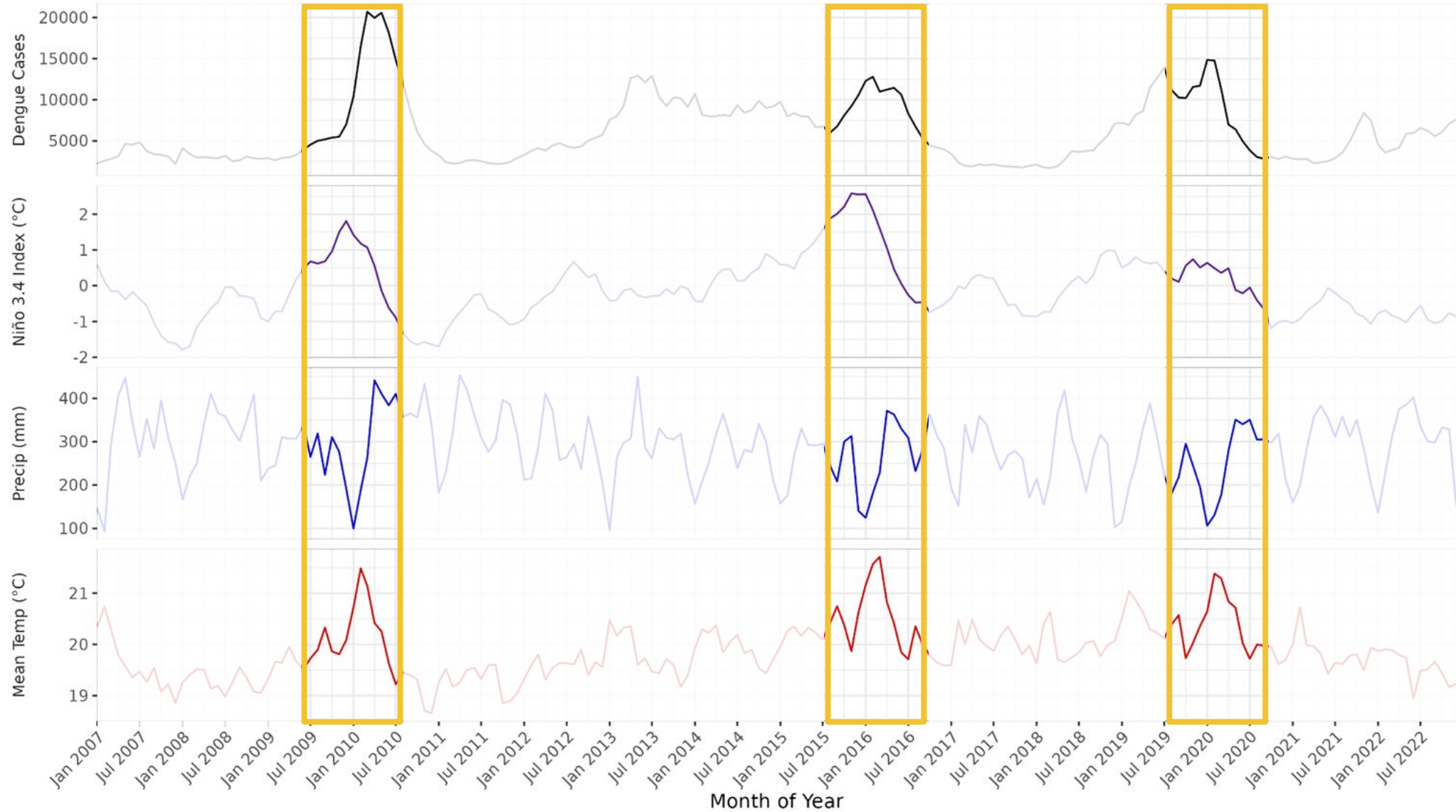
Data visualisation

Monthly Disease Cases, Mean Temperature, Precipitation and Niño 3.4 Index from Jan-07 to Dec-22

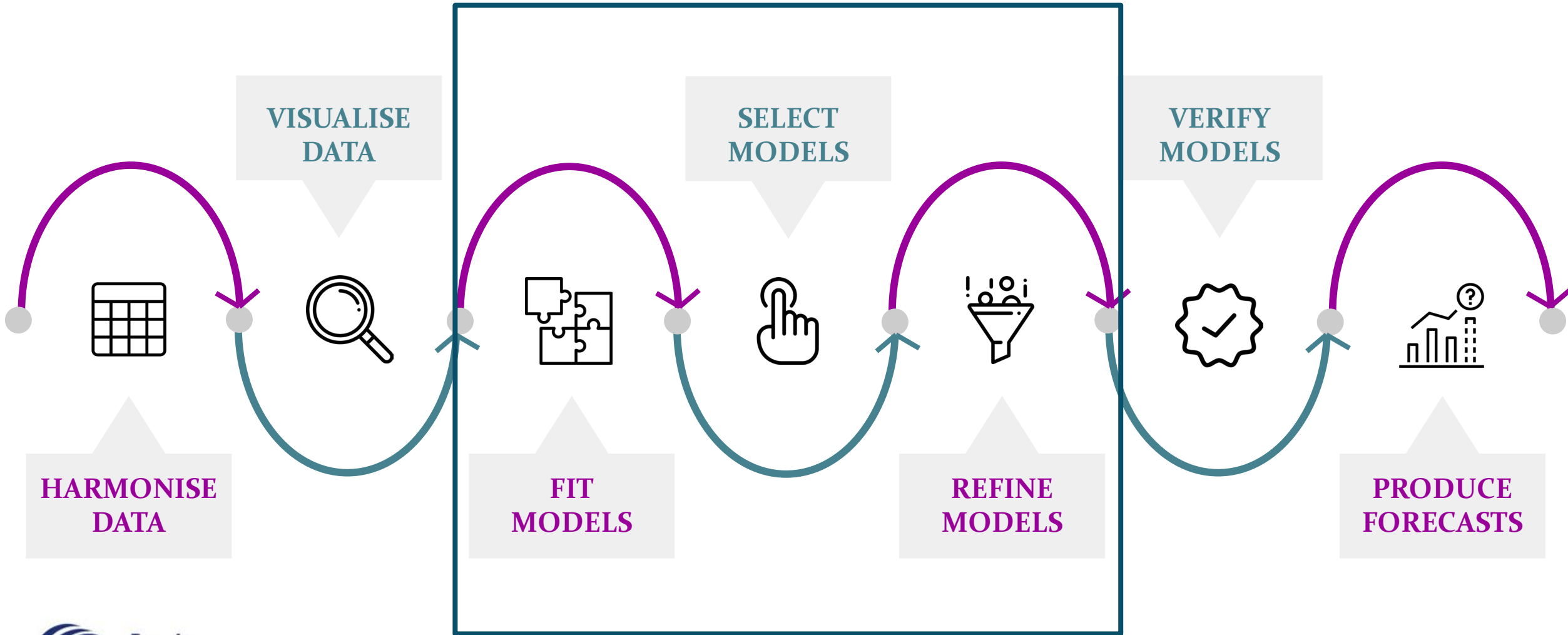


Data visualisation

Monthly Disease Cases, Mean Temperature, Precipitation and Niño 3.4 Index from Jan-07 to Dec-22

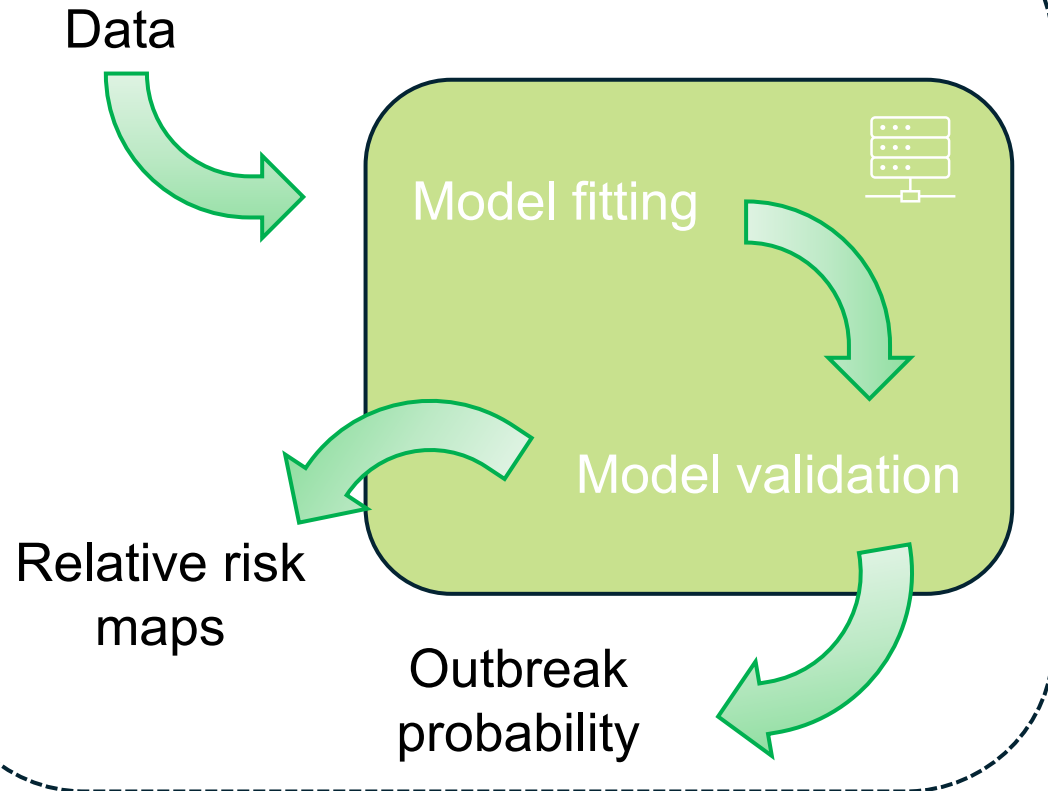


Risk modelling



Risk modelling

Bayesian ecological models

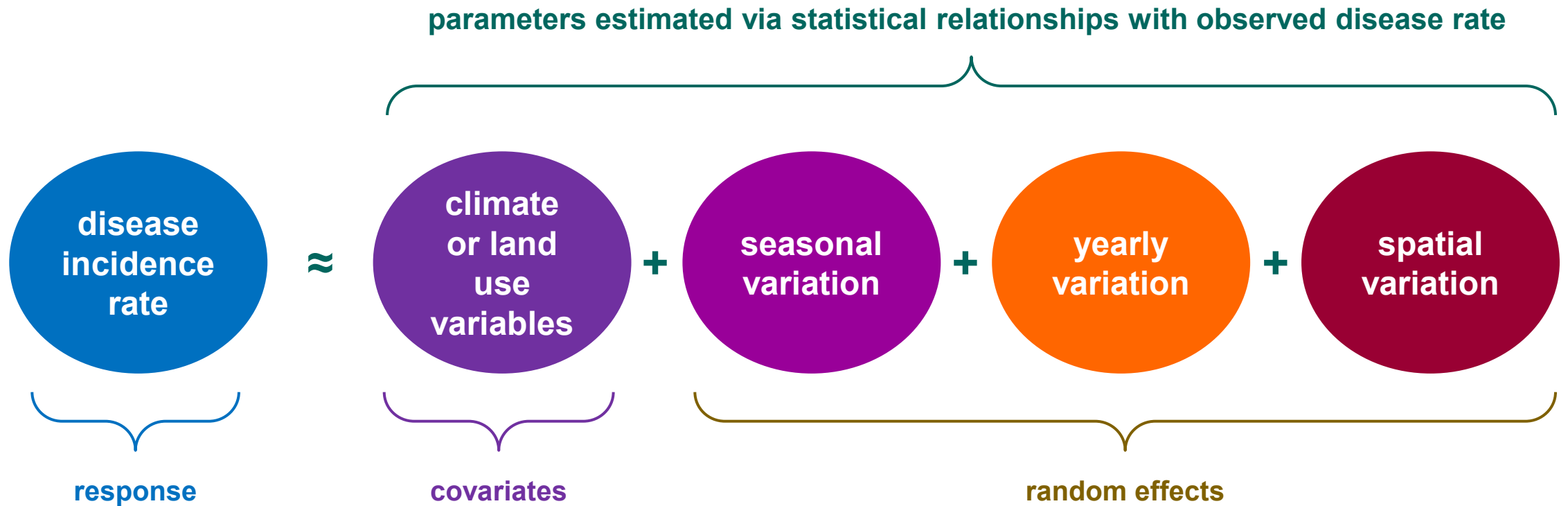


$$Y_{i,t} \mid \mu_{i,t}, \phi \sim \text{NegBin}(\mu_{i,t}, \phi)$$

$$\log(\mu_{i,t}) = \alpha + \log(P_{a[t,i]}) + \sum_k \beta_k X_{i,t,k} + \boxed{u_{a[i]} + v_{a[i]}} + \boxed{\delta_{m[t]} + \gamma_{a[t]}}$$

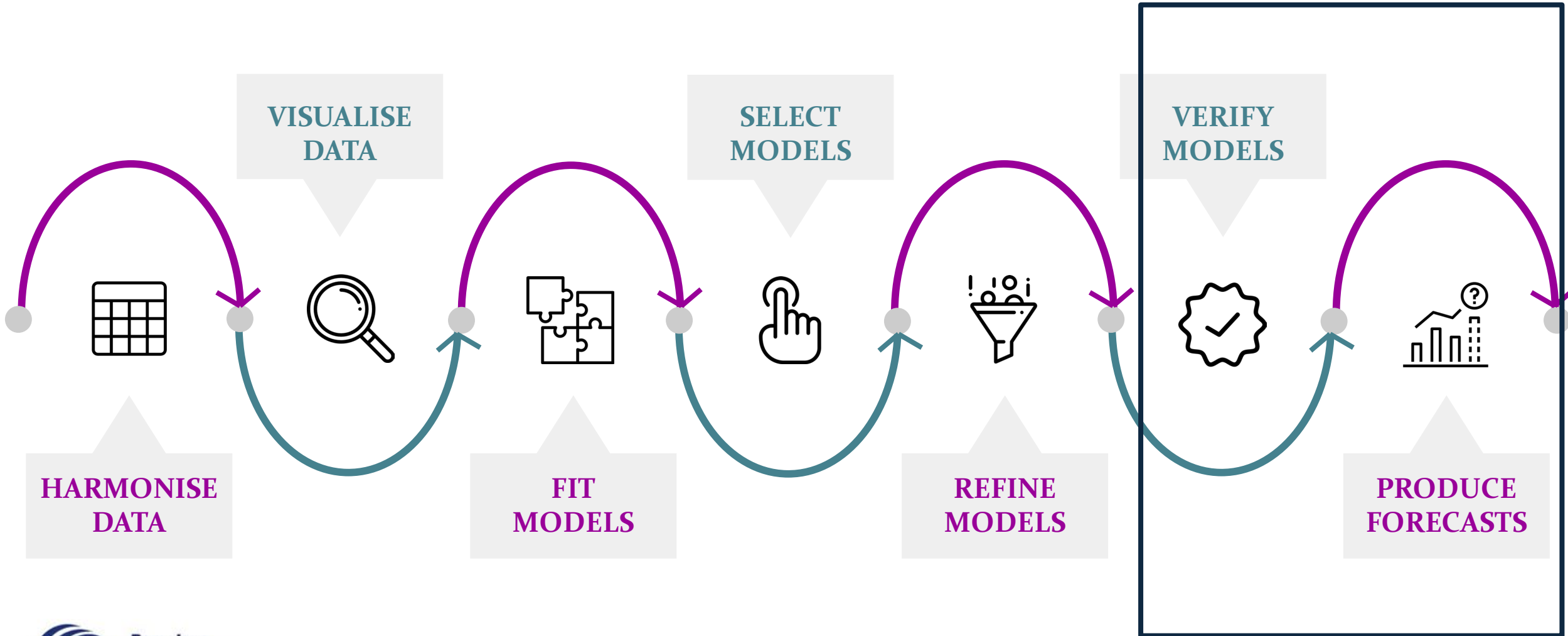
random effects

Risk modelling



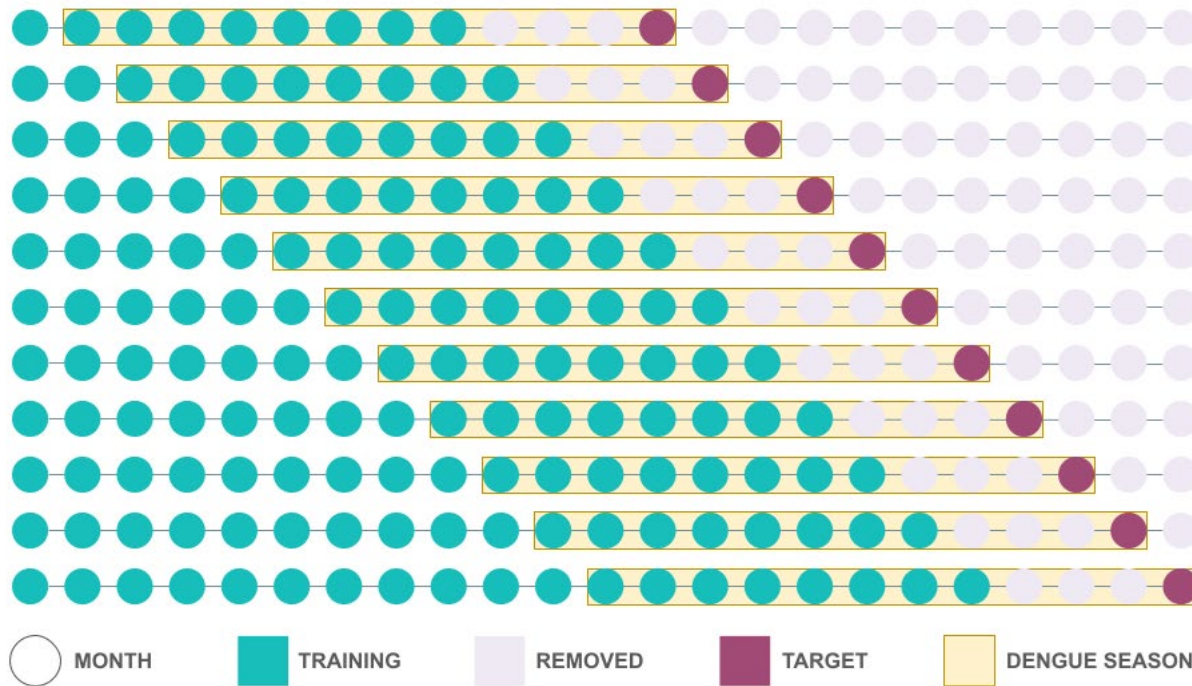
Fletcher et al. (in prep)

Risk modelling



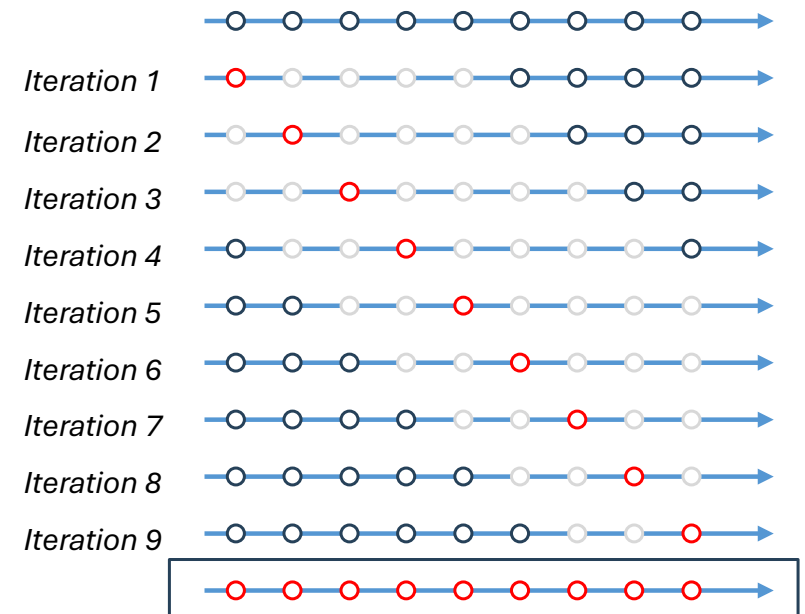
Model validation

Expanding window cross-validation

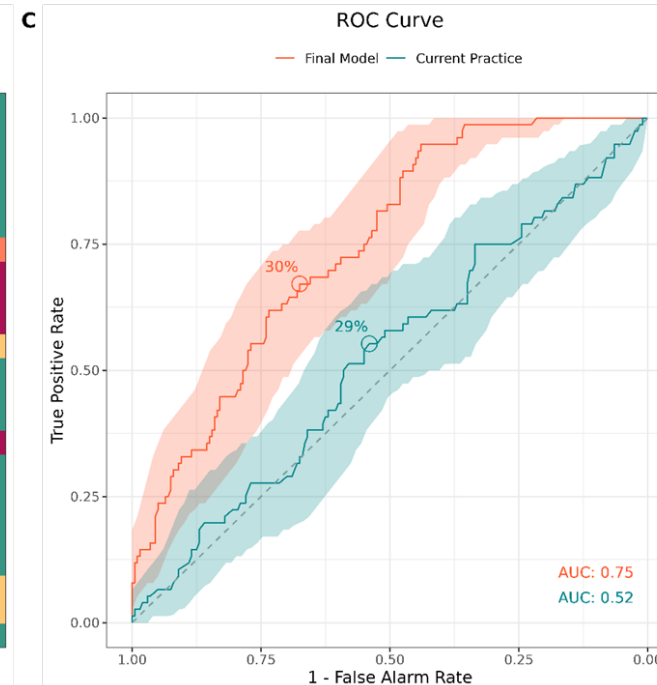
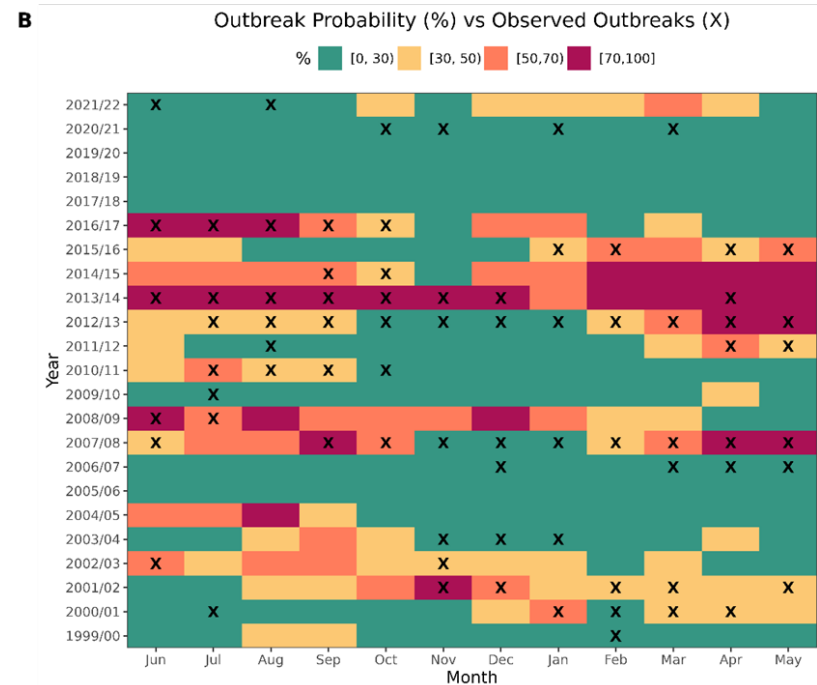
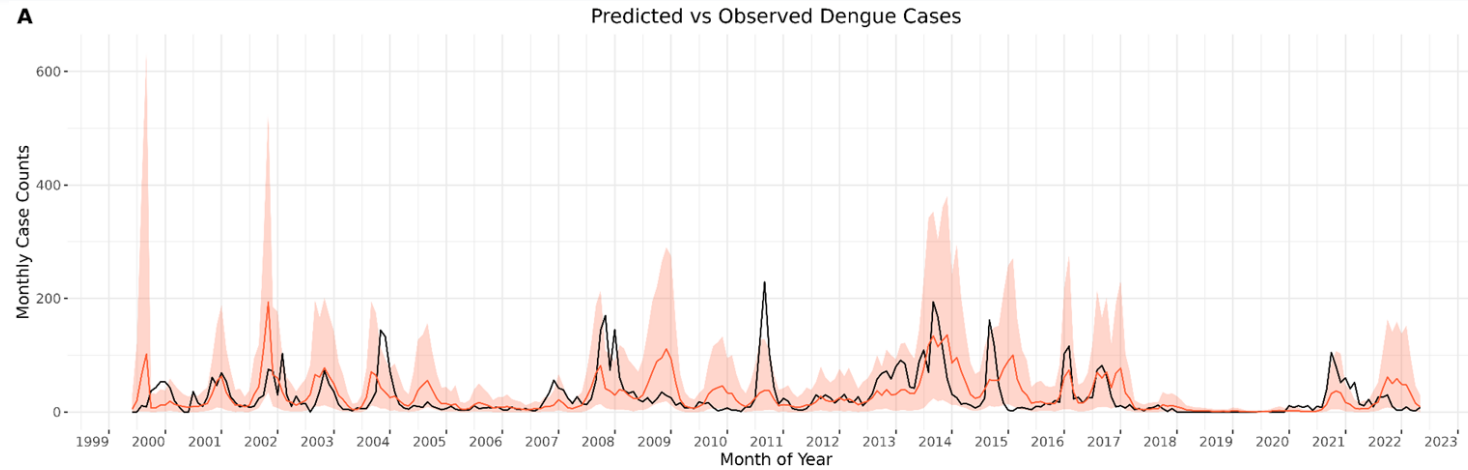


Fletcher et al. (2025)

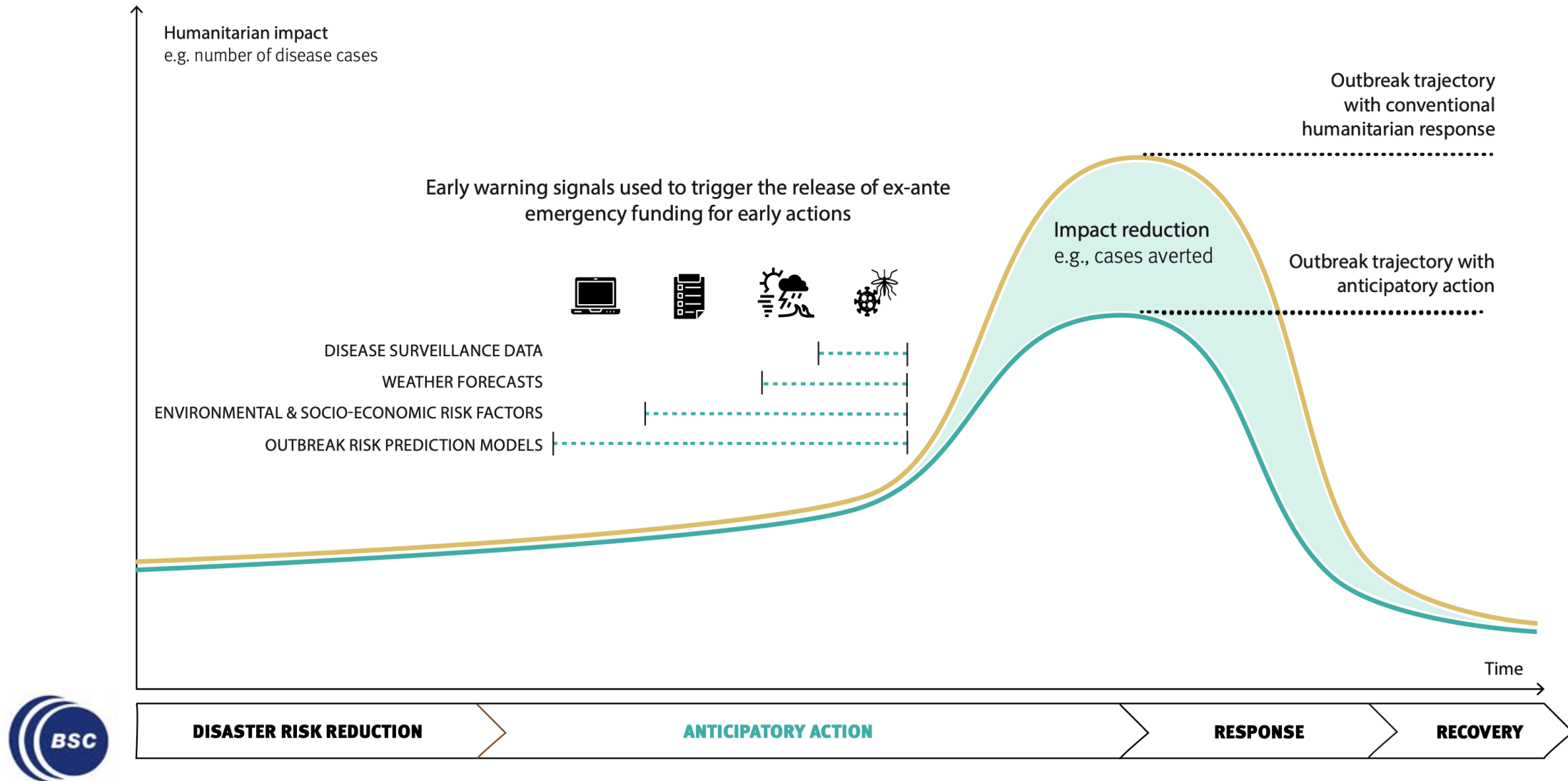
Leave n out cross-validation



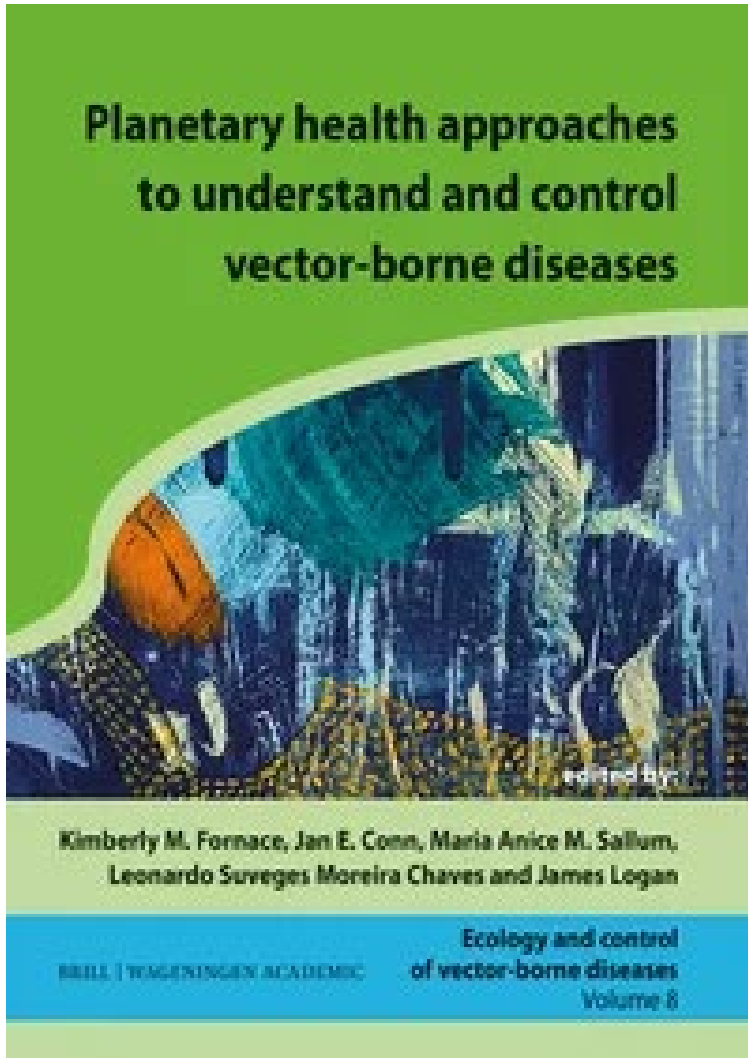
Model validation



Climate-informed anticipatory action



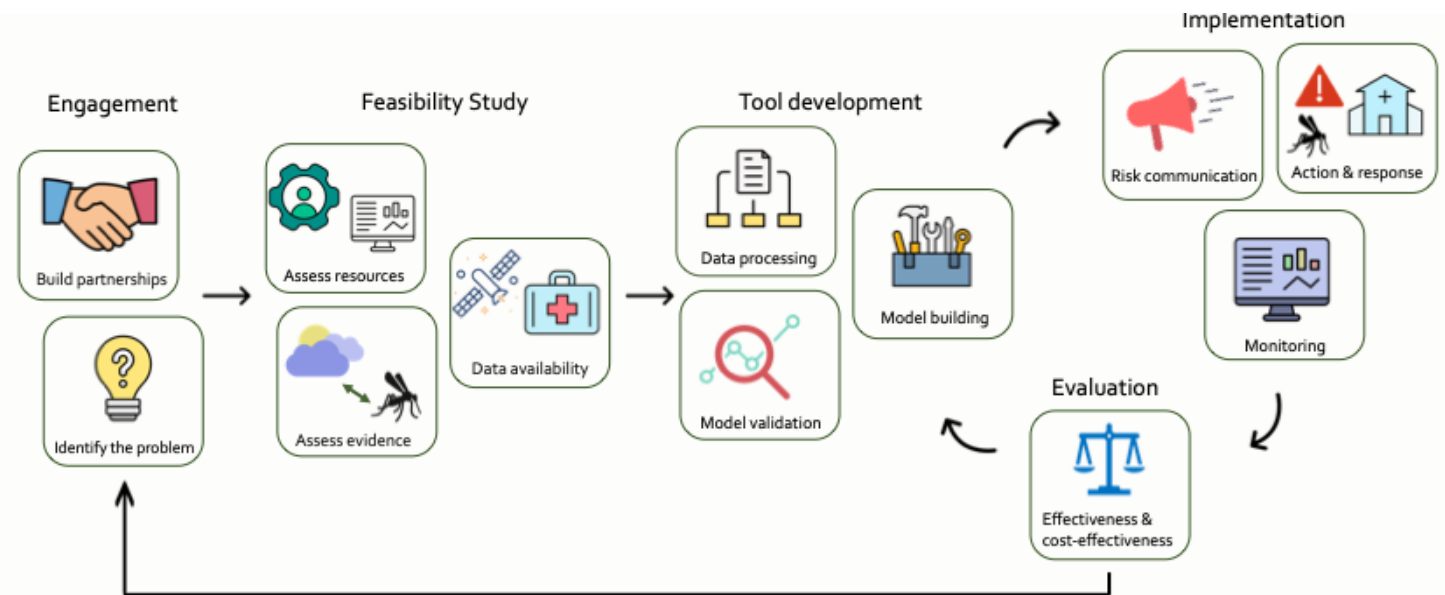
Early warning systems



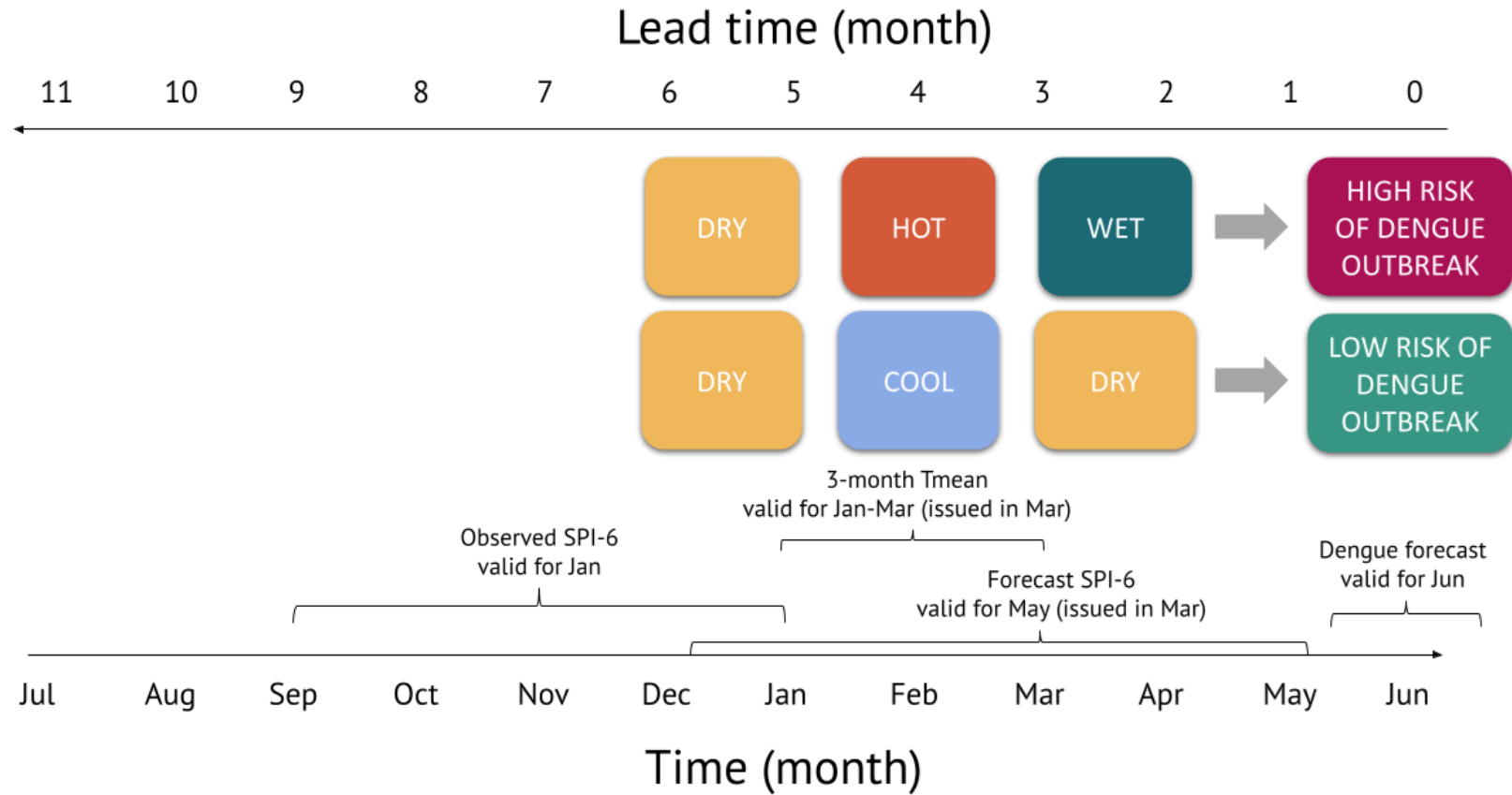
CHAPTER 12

Early warning systems for vector-borne diseases: engagement, methods and implementation

Emilie Finch^{1,2}, Martín Lotto Batista^{3,4}, Tilly Alcayna^{1,2,5}, Sophie A. Lee^{1,2},
Isabel K. Fletcher⁶ and Rachel Lowe^{1,2,4,7*}*



Early warning systems



Fletcher et al., (in prep)

Early warning systems



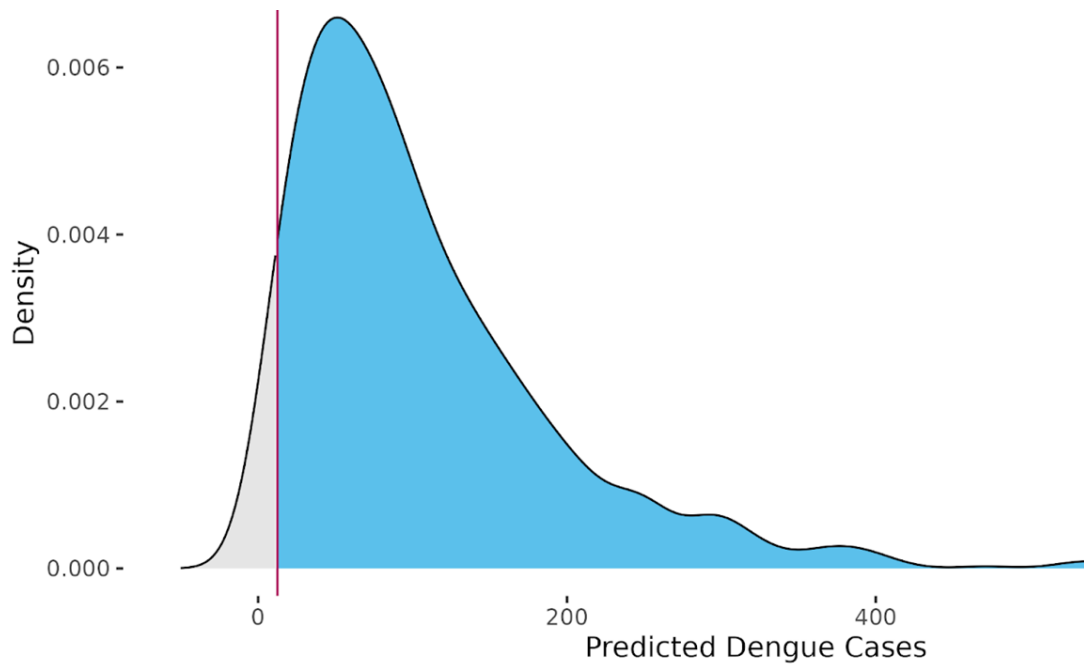
Caribbean Health Climatic Bulletin

Climate Conditions and Dengue in 2020

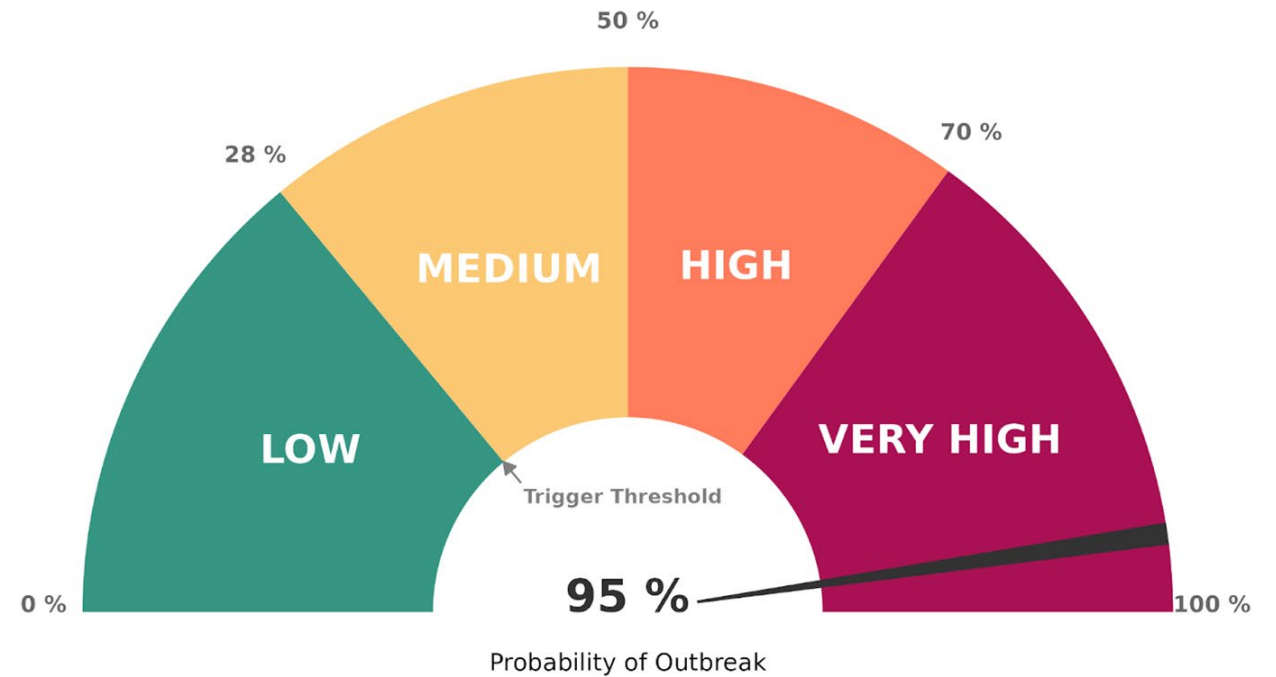
- Recent research (e.g. Lowe et al., 2018) on the link between climate conditions and dengue cases in eastern Caribbean countries suggests that drought conditions followed 4-5 months later by warmer than usual temperatures and excessive rainfall, increases the chance of Dengue outbreaks.
- In that regard, climate conditions in the Caribbean have been optimal for mosquito proliferation and dengue outbreaks throughout 2020, particularly in the eastern Caribbean. A regional drought implied increased water storage in the first half of 2020. This was followed by an intense heat season, particularly in the eastern Caribbean. Higher temperatures lead to increased rates of mosquito breeding, biting and disease transmission. The 2020 Caribbean wet season further brought episodes of excessive rainfall and flooding in many parts of the region, which contributed to an increase in mosquito breeding sites. Increased dengue case confirmations were recorded in several of the Eastern Caribbean states.

Early warning systems

Statistical Modeller

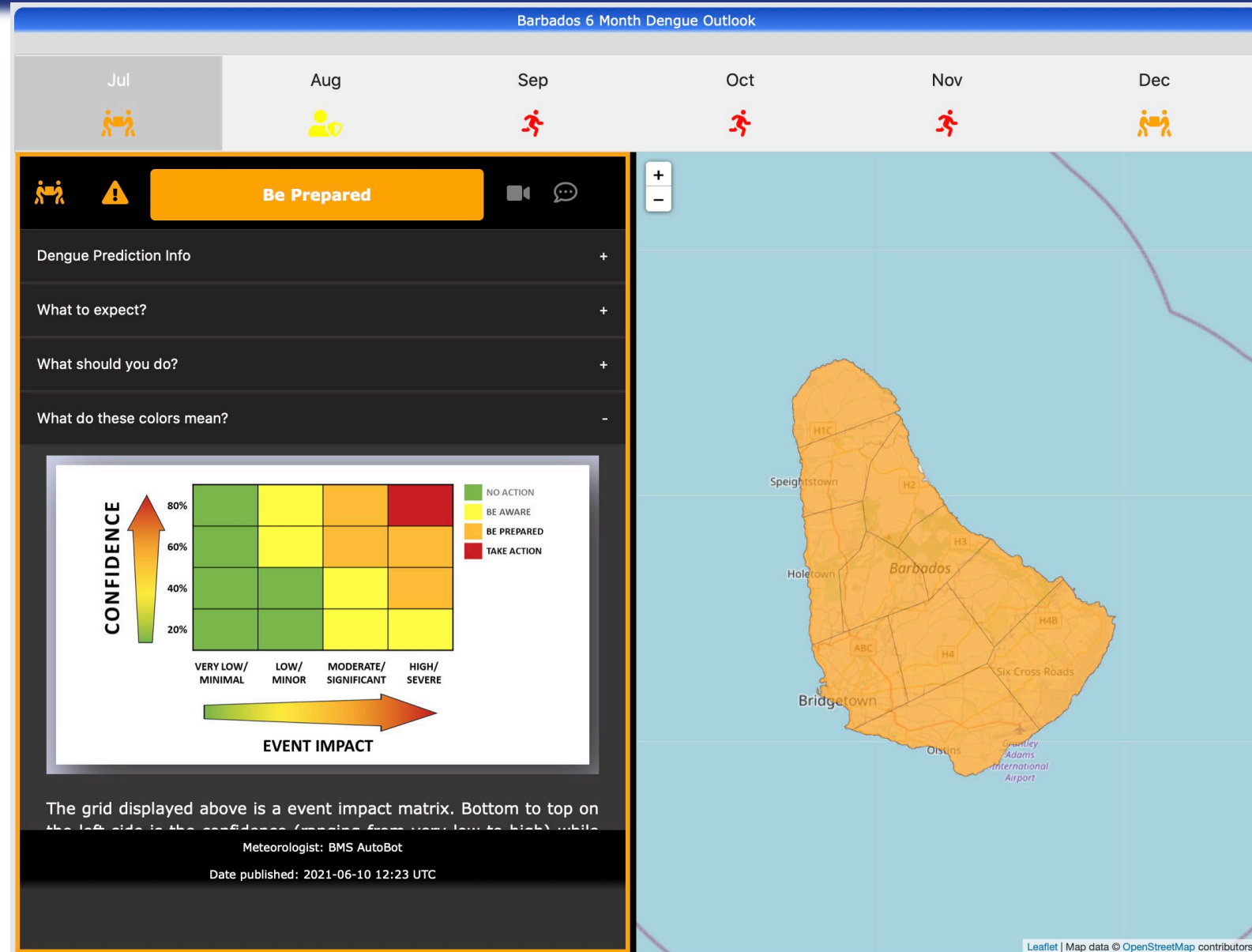


Chief Medical Officer



Translating probability to action

Prototype developed by
Barbados Meteorological Services



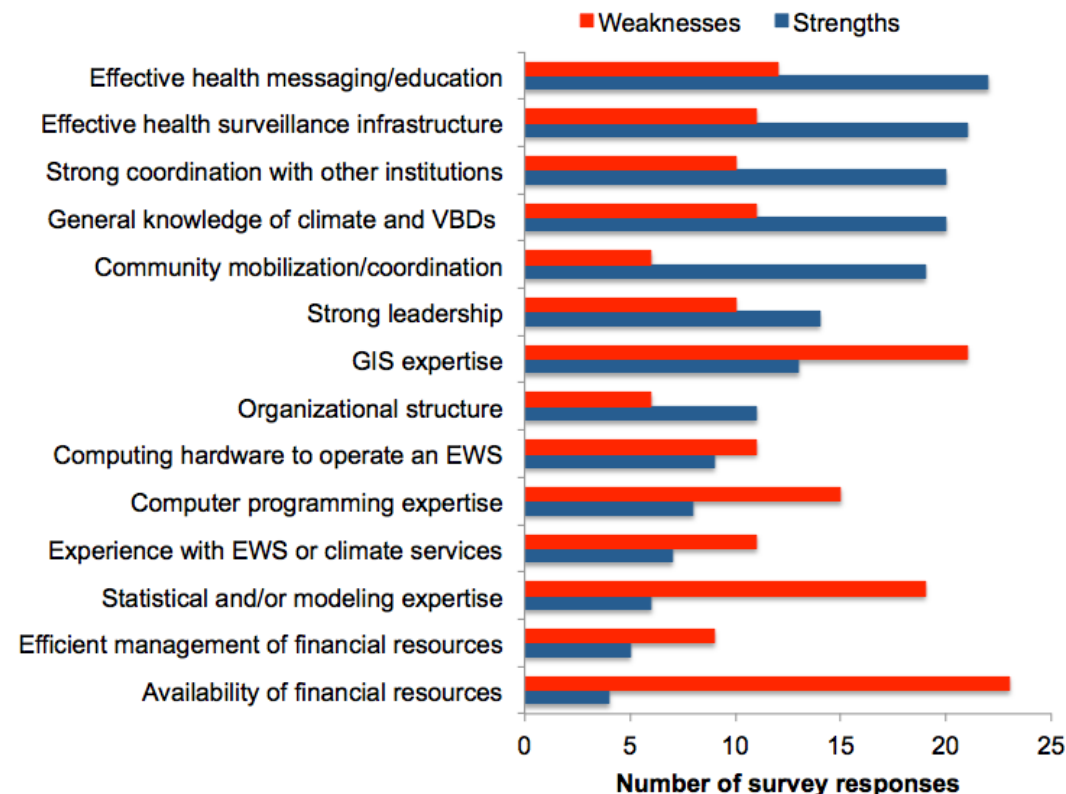
Technical capacity and financial resources



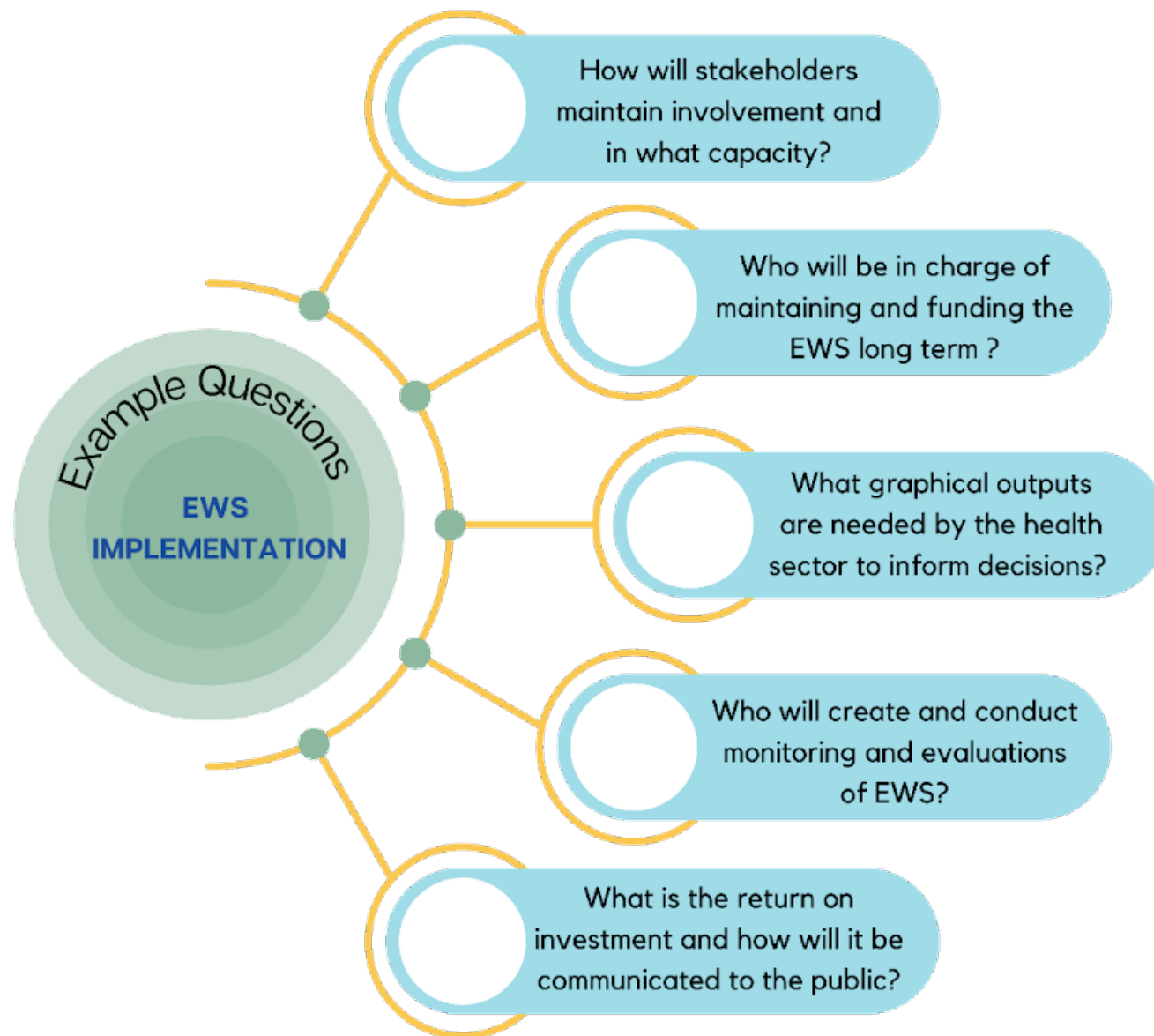
Caribbean Regional Climate Center



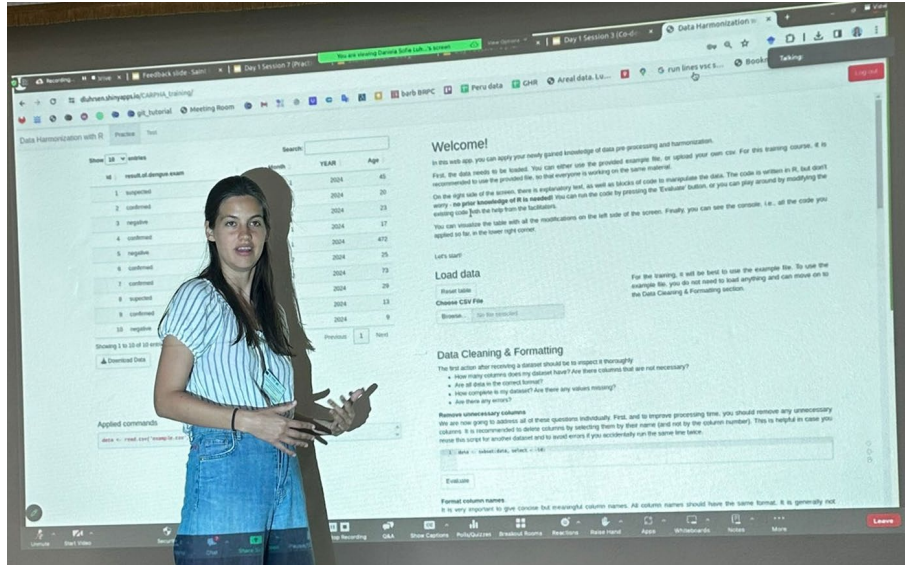
CARPHA



Guiding questions for EWS implementation



Capacity strengthening and knowledge exchange



Medical officer
of health



National
Dengue control



RDHS PDHS



Academics



Regional
Epidemiologist



Entomologist



Capacity strengthening and knowledge exchange

> [BMJ Glob Health](#). 2022 Jan;7(1):e007842. doi: 10.1136/bmjgh-2021-007842.

Co-learning during the co-creation of a dengue early warning system for the health sector in Barbados

Anna M Stewart-Ibarra ¹, Leslie Rollock ², Sabu Best ³, Tia Brown ³, Avriel R Diaz ⁴, Willy Dunbar ⁵, Catherine A Lippi ⁶, Roché Mahon ⁷, Sadie J Ryan ⁶, Adrian Trotman ⁷, Cedric J Van Meerbeeck ⁷, Rachel Lowe ^{8 9 10}

Affiliations + expand

PMID: 34992079 PMCID: [PMC8739055](#) DOI: [10.1136/bmjgh-2021-007842](#)

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RESEARCH ARTICLE

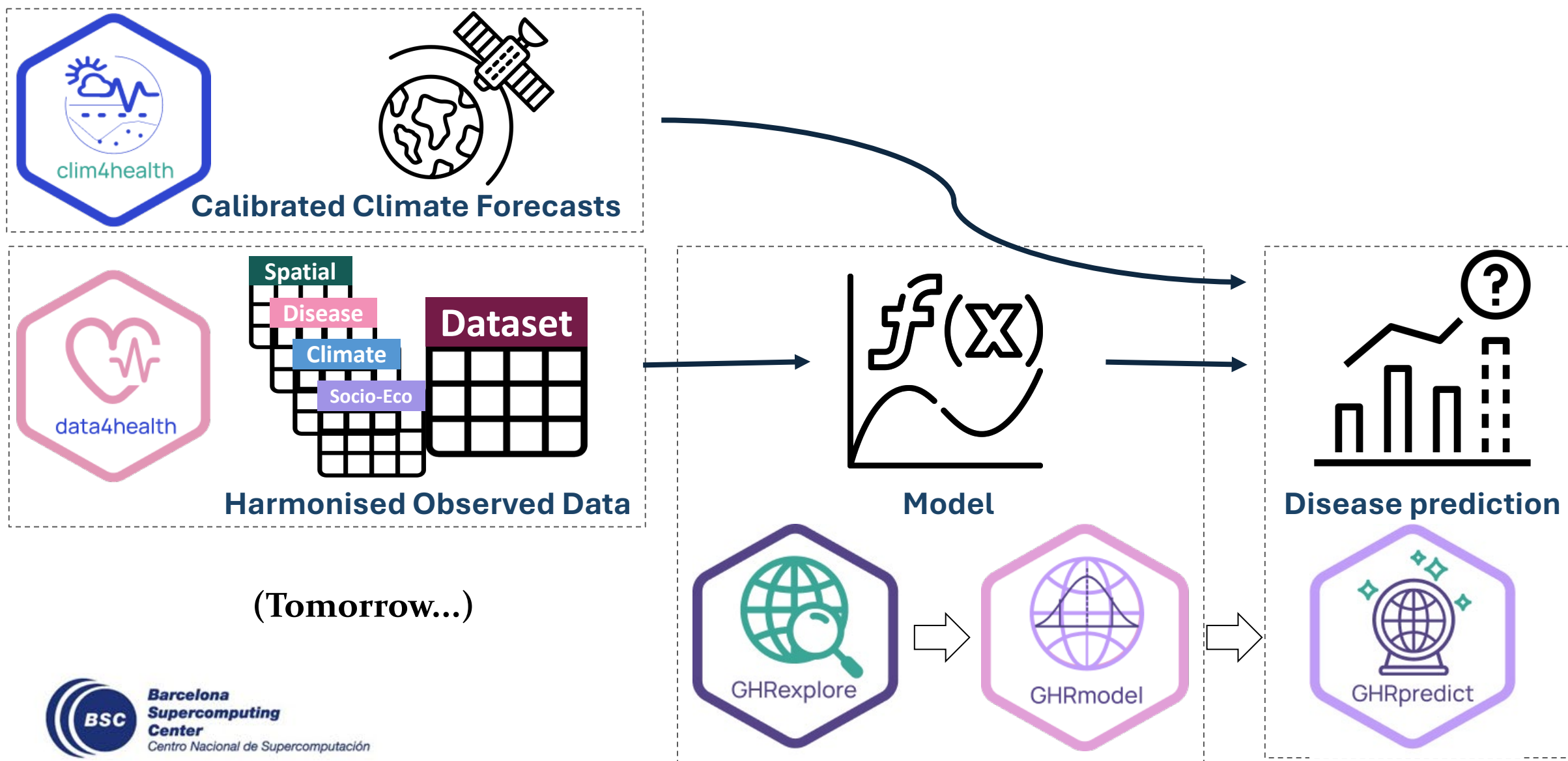
A demand-driven climate services for health implementation framework: A case study for climate-sensitive diseases in Caribbean Small Island Developing States

Avriel R. Díaz , Leslie Rollock, Laura-Lee G. Boodram, Roché Mahon, Sabu Best, Adrian Trotman, Cédric J. Van Meerbeeck, Chloe Fletcher, Willy Dunbar, Catherine A. Lippi, Daniela Lührsén, Cecilia Sorensen, Ángel G. Muñoz, Sadie J. Ryan, Anna M. Stewart-Ibarra, Rachel Lowe [[view less](#)]

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Capacity strengthening and knowledge exchange



Thank you

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